

Student Performance Analysis: Exploratory and Predictive Modeling

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1 PART A: Exploratory Data Analysis

This report presents a comprehensive exploratory analysis of the “Student Performance” dataset from the UCI Machine Learning Repository. The dataset contains information on 395 students from Portuguese secondary schools, encompassing demographic characteristics, family background, academic history, and final performance metrics. Our analysis focuses on identifying key associations with student academic achievement through well-structured research questions and appropriate statistical visualization techniques.

1.1 Research Question 1: Parental Education and Grades

Question: How is maximum parental education associated with mean final grade (G3)?

Methodology: We created a composite variable representing the higher educational attainment between mother (Medu) and father (Fedu), as a descriptive summary of the two parental education measurements. A bar chart with error bars (standard error of the mean) visualizes the relationship between parental education levels and student performance.

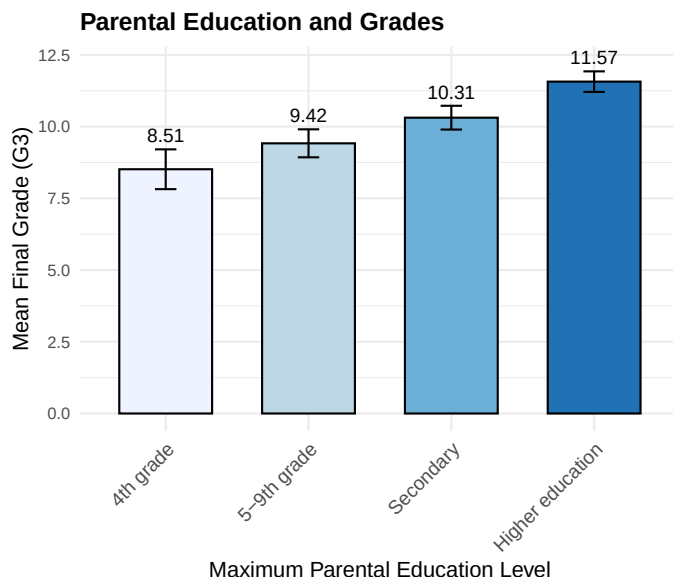


Figure 1: Maximum Parental Education and Student Final Grade

Interpretation: Mean final grade increases across these parental-education groups, from 8.51 to 11.57. The bars show mean plus/minus one standard error, not confidence intervals or a hypothesis test. These unadjusted observational associations do not establish a causal mechanism.

1.2 Research Question 2: Grade Distribution Analysis

Question: What is the distribution of final grades (G3) relative to the passing threshold (10)?

Methodology: We transformed the final grades by subtracting the passing threshold (10 points) to create a “distance-to-pass” variable. This approach provides immediate insight into the proportion of students above, at, or below the critical performance benchmark. A histogram with color-coded regions effectively communicates the distribution patterns.

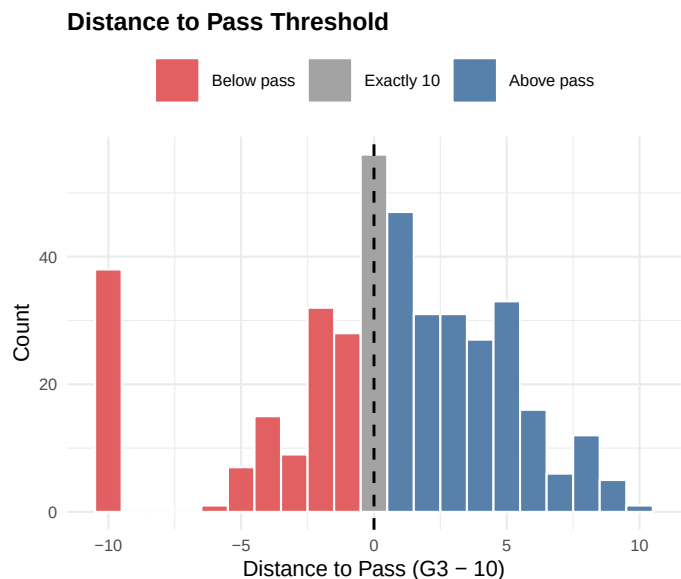


Figure 2: Distribution of Final Grades Relative to Passing Threshold

Interpretation: 67.1% of students pass and 32.9% fail. A recorded zero does not by itself identify dropout or exam absence. The figure describes outcomes; it does not estimate an intervention effect.

1.3 Research Question 3: Educational Aspirations and Grades

Question: How does a student's aspiration for higher education relate to their final grade (G3) distribution?

Methodology: We employed a combination visualization (violin plot overlaid with boxplot) to examine both the distribution shape and central tendencies of grades by educational aspiration. This approach reveals not only differences in medians but also the entire distribution characteristics, including potential multimodality or skewness patterns.

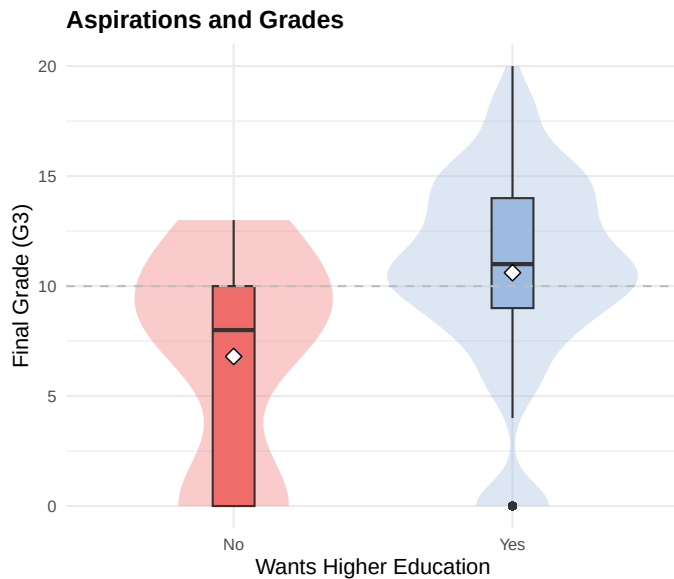


Figure 3: Final Grades by Higher Education Aspiration

Interpretation: Pass rates are 68.8% among students reporting higher-education aspirations and 35.0% among those not reporting them. The groups differ in other observed and unobserved characteristics, so the association cannot establish whether aspirations change grades.

1.4 Research Question 4: Academic History Associations

Question: How do the observed G2 and G3 values vary across groups with and without past academic failures?

Methodology: A scatter plot colors second-period (G2) and final grades (G3) by failure history and overlays a pooled regression line. It describes the sample relationship without testing differences in slopes between groups.

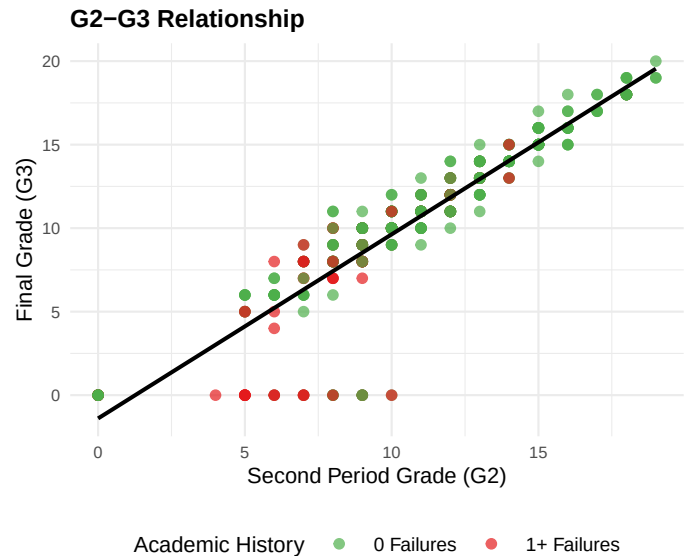


Figure 4: Relationship between G2 and G3 by Academic Failure History

Interpretation: The Pearson correlation between G2 and G3 is 0.905. The black regression line is pooled across failure-history groups. It does not test a difference in slopes or establish the effect of past failure.

1.5 Research Question 5: Rank Correlation Analysis

Question: Which numerical/ordinal factors have the strongest correlation with final grade (G3)?

Methodology: We calculated Spearman rank correlations between the selected numeric/ordinal variables and G3, presenting results via a lollipop plot. Rank correlations describe marginal monotonic associations; they do not measure adjusted predictive importance.

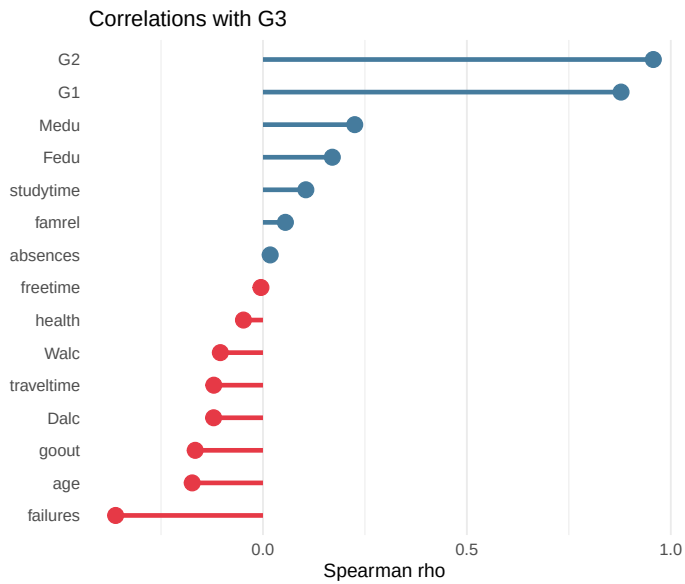


Figure 5: Spearman Correlation of Factors with Final Grade

Interpretation: The strongest Spearman correlations with G3 are G2 (0.957) and G1 (0.878). Past failures correlate negatively (-0.361), whereas absences are close to uncorrelated (0.018). These marginal associations are not causal effects or adjusted predictive importances.

2 PART B: Predictive Decision Tree Analysis

2.1 Objective and Methodology

This section evaluates a decision tree model to predict whether students achieve satisfactory performance ($G3 \geq 10$) using demographic, social, and academic variables while excluding prior grades ($G1$, $G2$) to examine prediction without earlier grade measurements.

2.2 Model Development Process

Data Preparation: We created a binary target variable classifying students as passing ($G3 \geq 10$) or failing ($G3 < 10$). The dataset was split into 70% training (277 observations) and 30% testing (118 observations) using stratified sampling to maintain class proportions.

Model Training: We employed 10-fold cross-validation with ROC optimization to tune the complexity parameter (cp), testing 34 different values from 0.001 to 0.1. Cross-validation selects among the specified candidates; it does not guarantee prevention of overfitting.

2.3 Optimized Decision Tree Structure

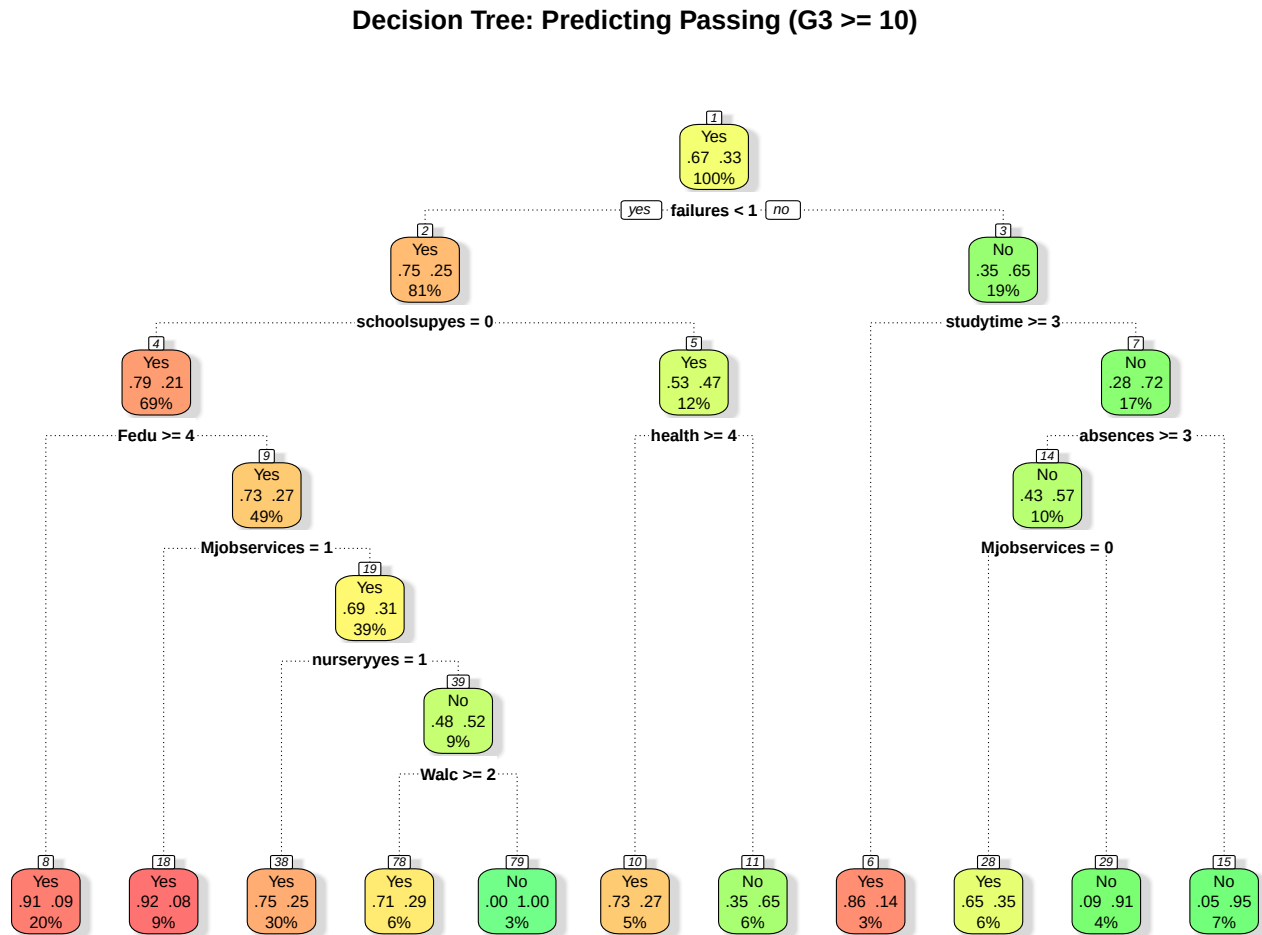


Figure 6: Optimized Decision Tree for Predicting Passing

Tree interpretation: Read the fitted splits and leaf probabilities from the diagram. They describe associations in this training sample; leaf probabilities are not calibrated individual risk estimates, and split order is not a causal ranking.

2.4 Model Performance Evaluation

Performance Metrics on Test Set:

- **Accuracy:** 0.5932
- **Kappa:** -0.0408 (agreement beyond chance)
- **Sensitivity (Recall):** 0.8101
- **Specificity:** 0.1538
- **Precision (Positive Predictive Value):** 0.6598
- **F1-Score:** 0.7273 (balance of precision and recall)

Performance Analysis: Test accuracy is 0.5932, compared with 0.6695 for always predicting the training-majority class. Specificity is recall for failing students: 0.1538. These results do not support describing this model as an effective early-warning system. Accuracy alone also does not measure probability ranking or calibration.

2.5 Conclusions and limitations

This analysis describes one sample of 395 students and evaluates a tree on 118 held-out observations. Cross-validation selects the complexity parameter using the training partition only. The outcome means passing, not exceptional academic performance.

The descriptive associations do not identify causes or demonstrate that an intervention would improve outcomes. Excluding G1 and G2 is insufficient to establish an early-warning use case: the time at which remaining variables, including accumulated absences, become available would also need verification.

The held-out results must be compared with the majority-class baseline and the poor recall for failing students. Independent schools or cohorts, probability calibration, uncertainty estimates and an assessment of intervention costs would be required before considering educational use.

Dataset: UCI Student Performance, mathematics subset, Cortez and Silva (2008).